

Intelligent Placement Management System: A Role-Based Web Platform for Automated Eligibility Matching, AI-Assisted Assessment, and On-Device Proctoring

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Abstract—Campus recruitment at higher-education institutions is predominantly executed through disconnected spreadsheets, unmonitored messaging channels, and manual academic verification, leading to administrative overhead, human auditing oversights, and scheduling latency. This paper presents the design, technical architecture, and empirical verification of the **Intelligent Placement Management System (IPMS Elite)**, a production-grade, role-based web ecosystem unifying students, institutional placement administrators, and corporate recruiters. The system implements: (1) client-evaluated reactive eligibility screening across multi-parameter academic constraints, (2) automated multimodal vision-language verification of student marks cards and resume tamper screening, (3) an on-device, privacy-preserving examination proctoring engine executing quantized MobileNet-SSD models over WebGL without streaming video off-device, (4) an end-to-end recruitment drive management pipeline supporting multi-round evaluations and verified Google OAuth corporate onboarding, and (5) role-contextual conversational intelligence powered by Google Gemini 2.5 Flash. The platform is constructed on a React 18, TypeScript, and Supabase PostgreSQL architecture safeguarded by declarative Row-Level Security policies. Empirical benchmarks demonstrate an on-device proctoring detection sensitivity of 97.8% at a false positive rate of 1.2% using a temporal persistence filter ($\$k_{\{frames\}}=3\$$), document extraction accuracy of 98.3%, and a mean System Usability Scale (SUS) benchmark of 86.4/100 ("Grade A") across student and recruiter cohorts.

Index Terms—Campus placement management, on-device machine learning, online examination proctoring, multimodal document verification, dynamic eligibility automation, Row-Level Security, web application architecture.

I. INTRODUCTION

Campus recruitment drives represent one of the most critical annual operations conducted by engineering institutions, bridging academic cohorts with corporate employers. During a typical recruitment cycle, an institutional Training and Placement Cell (T&P) must continuously maintain and verify the credentials of hundreds of candidates, negotiate job criteria with dozens of visiting enterprises, organize multi-round competitive examinations, enforce test integrity, and coordinate shortlisting rosters.

In conventional institutional practice, this operational pipeline is fragmented across isolated desktop spreadsheets, shared network drives, email threads, and third-party communication portals [1], [2]. This ad hoc paradigm introduces several severe systemic vulnerabilities:

- **Eligibility Verification Latency:** Student academic records (CGPA, active backlogs, cleared courses) are cross-referenced manually against company cutoffs, consuming dozens of coordinator hours and inviting human error [3].
- **Data Vulnerability and Falsification:** Unverified, self-reported candidate credentials allow GPA inflation and backlog omissions to pass undetected until late verification stages.
- **Surveillance Overreach vs. Integrity Dilemma:** Remote examinations are either conducted unproctored or depend on commercial cloud surveillance tools that continuously stream raw video to remote third-party servers. This model raises severe student privacy anxieties, bandwidth constraints, and opaque algorithmic flags [6].
- **Opaque Evaluation and Feedback Deprivation:** Candidates typically receive an arbitrary pass/fail result with zero diagnostic explanation of misconceptions or subject weaknesses.

To resolve these operational deficits, we engineered the **Intelligent Placement Management System (IPMS)**, a unified, role-governed web platform deployed in collaboration with the Training and Placement Cell of Malnad College of Engineering, Hassan. The architecture adheres to three primary design tenets:

- **Relational Single Source of Truth:** Every student profile, company drive, test attempt, and administrative action is preserved within a unified PostgreSQL database governed by declarative Row-Level Security (RLS).
- **Automation of Routine Verification:** Eligibility checking, academic marks card OCR, question authoring, proctoring, and scoring are delegated to local and serverless routines.
- **Transparent Human-in-the-Loop Governance:** Automated algorithmic flags (such as proctoring warnings or shortlisting scores) are inspectable, tunable, and reversible by placement officers.

II. RELATED WORK

Early campus placement management software focused primarily on digitizing student resumes into basic searchable

relational databases [2], [11]. Subsequent systems introduced rule engines to filter candidates against static database flags [1], [3]. A fundamental flaw in this paradigm is that eligibility flags were pre-computed in periodic batch jobs; any late-breaking grade revision or backlog clearance was not reflected until the batch script re-executed, causing qualified students to be excluded from scheduled drives.

In document verification, traditional Optical Character Recognition (OCR) systems such as Tesseract [10] perform reasonably on high-contrast uniform text, but struggle severely on photographed institutional marks cards containing tabular borders, complex watermarks, and perspective skew. The advent of vision-language foundation models, exemplified by the Google Gemini model family [9], enables joint visual-semantic document reasoning capable of parsing irregular tabular structures while detecting typographic anomalies.

In the examination integrity domain, prior automated proctoring frameworks predominantly utilize cloud-streaming architectures [4], [5]. These systems transmit continuous WebRTC video streams to cloud servers for computer vision inference. In addition to high upstream bandwidth demands, storing biometric video recordings on commercial servers poses ethical and data-compliance hazards [6]. In contrast, browser-native machine learning runtimes such as TensorFlow.js [7] enable deep convolutional object detection models (e.g., MobileNet-SSD [8]) to execute directly on client hardware via WebGL. Table I summarizes IPMS's design innovations over conventional approaches.

TABLE I
IPMS RELATIVE TO CONVENTIONAL PLACEMENT SYSTEMS

Operational Dimension	Conventional Method	IPMS Approach
Eligibility Screening	Static batch jobs / manual spreadsheet review	Reactive, client-evaluated on profile fetch (<5 ms)
Marks Verification	Manual physical check / template OCR	Multimodal vision model with tamper screening
Test Invigilation	Cloud video streaming or unproctored	On-device WebGL inference; zero video transmitted
Candidate Feedback	Numeric score only	Generative AI per-question diagnostic analysis
Recruiter Onboarding	Manual coordination / static forms	Verified Google OAuth & AI JD document parsing
Audit & Security	Ad-hoc logs / none	Database RLS, TOTP 2FA, immutable audit trail

III. SYSTEM ARCHITECTURE

IPMS is designed as a cloud-synchronized Single Page Application (SPA) leveraging a modular four-tier architecture:

1) Presentation Tier: Developed in React 18 and TypeScript with Vite 5. UI components utilize Tailwind CSS and shaden/ui primitives. Animations are orchestrated via Framer Motion.

2) State and Validation Tier: Remote data caching and optimistic UI updates are managed via TanStack React Query v5. Form state is handled by React Hook Form validated against strict runtime Zod schemas.

3) Serverless Intelligence Tier: Serverless Edge Functions handle sensitive operations: Google Gemini 2.5 Flash API calls, PDF parsing, generative diagnostic feedback, and transactional Gmail SMTP email dispatch via a dedicated Vercel edge relay.

4) Persistence and Access Control Tier: A Supabase PostgreSQL instance enforces Row-Level Security (RLS) on every table. Candidate resumes and marks cards reside in private storage buckets accessible solely via short-lived signed URLs. Table II details the technology stack.

TABLE II
COMPLETE SYSTEM TECHNOLOGY STACK

System Layer	Technology	Functional Responsibility
Language	TypeScript 5.8	End-to-end static type safety
UI Framework	React 18.3 + Vite 5.4	Reactive client application
Styling & 3D	Tailwind CSS + shaden/ui	Adaptive theme system
Server State	TanStack Query v5	Client-side caching & invalidation
Database	PostgreSQL 15 (Supabase)	Relational persistence & RLS
AI Engine	Google Gemini 2.5 Flash	JD parsing, question synthesis, TTS/STT
Edge ML	TensorFlow.js + MobileNet-SSD	In-browser gadget detection
Authentication	Supabase Auth + RFC-6238	Google OAuth & TOTP 2FA
Messaging	Nodemailer (Gmail SMTP)	Transactional recruiter invitations
Unit Testing	Vitest + Testing Library	Automated test suite (57 tests)

IV. DATABASE DESIGN AND SECURITY ARCHITECTURE

The persistence model comprises ten relational tables: `profiles`, `user_roles`, `companies`, `drive_rounds`, `tests`, `test_attempts`, `schedules`, `notifications`, `audit_logs`, and `company_invites`.

Authorization is decoupled from profile data: user roles are segregated into an independent `user_roles` table rather than a mutable role column on `profiles`. This mitigates client-side privilege escalation. A `SECURITY DEFINER` helper function prevents circular RLS policy recursion:

```
CREATE OR REPLACE FUNCTION public.has_role(
  _user_id UUID, _role app_role)
RETURNS BOOLEAN LANGUAGE sql STABLE SECURITY DEFINER
SET search_path = public AS $$
  SELECT EXISTS (
    SELECT 1 FROM public.user_roles
    WHERE user_id = _user_id AND role = _role
  );
$$;
```

Every public table is governed by RLS policies. Students are constrained to reading and writing records matching their authenticated `auth.uid()`, while administrative access is validated via `has_role(auth.uid(), 'admin')`.

V. CORE FUNCTIONAL MODULES & IMPLEMENTATION

A. Multi-Role Authentication, Google OAuth, and Session Governance

Candidate registration supports email/password or Google OAuth. To protect institutional operations, administrator and recruiter accounts cannot self-register; they require single-use, cryptographically generated tokens stored in `company_invites`.

Administrator accounts require Time-based One-Time Password (TOTP) two-factor authentication, even following successful Google OAuth. Student sessions are tied to browser tab lifecycle, mitigating credential leakage in shared university computer labs. Mutating administrative operations are written synchronously to `audit_logs`.

B. Verified Recruiter Onboarding & Document AI Extraction

Visiting recruiters receive a tokenized link: `/login?company_invite=${token}&role=company`. When authenticating with Google, the client preserves the invite token in `localStorage` across OAuth redirects. Upon first login, the recruiter is intercepted by `ProtectedRoute` and redirected to `/company/onboarding`.

Recruiters can upload their placement brochure or Job Description (JD) in PDF, DOCX, or TXT format. An edge-resident Gemini 2.5 Flash multimodal extraction prompt parses company background, job designations, salary breakdown (Fixed LPA, Variable LPA, Total CTC), eligibility cutoffs, and selection rounds. Any missing parameters trigger an alert banner highlighting fields for manual completion before campus activation. Recruiters retain full edit access via `/company/onboarding?mode=edit`.

C. Dynamic Multi-Parameter Eligibility Matching Engine

Eligibility is evaluated reactively on the client. Let candidate profile $S = \langle \text{CGPA}_S, \text{Branch}_S, \text{Backlogs}_S, K_S, \text{Sem}_S \rangle$, where K_S represents verified technical skills. Let company drive criteria $C = \langle \text{CGPA}_C^{\min}, B^C, \text{Backlogs}_C^{\max}, K^C, S^C \rangle$. The eligibility predicate $E(S, C) \in \{0, 1\}$ is formulated as:

$$E(S, C) = \mathbb{I} \left(\text{Big}(\text{CGPA}_S \geq \text{CGPA}_C^{\min}) \wedge (\text{Branch}_S \in B^C) \wedge (\text{Backlogs}_S \leq \text{Backlogs}_C^{\max}) \wedge (K^C \subseteq K_S) \wedge (\text{Sem}_S \in S^C) \right) \quad (1)$$

Because TanStack Query maintains profile data in memory, eligibility updates instantly when a new grade is verified, requiring zero database batch writes.

D. Academic Document Verification & CGPA Aggregation

Marks cards are uploaded to private storage and forwarded to a serverless vision model. The model extracts University Seat Number (USN), student name, semester index, and SGPA, screening for typography inconsistencies. Upon successful validation, aggregate CGPA is updated automatically:

$$\text{CGPA} = \frac{1}{N} \sum_{i=1}^N \text{SGPA}_i \quad (2)$$

This eliminates self-reported grade misrepresentation.

E. Multi-Round Recruitment Drive Pipeline

Recruiters configure sequential selection rounds (Online Assessments, Technical Interviews, Group Discussions, HR Fitment) in `drive_rounds`. Candidates progress through a state-machine lifecycle: $\text{pending} \rightarrow \text{qualified} \rightarrow \text{not_qualified} \rightarrow \text{absent}$. Candidate shortlisting utilizes a composite score balancing test results, academic consistency, and improvement:

$$S_{\text{shortlist}} = 0.40 \cdot S_{\text{test}} + 0.30 \cdot \text{CGPA}_{\text{scaled}} + 0.30 \cdot \Delta_{\text{improvement}} \quad (3)$$

where $\Delta_{\text{improvement}} = \max(0, S_{\text{test}}^{\text{Attempt}} - S_{\text{test}}^{\text{Attempt-1}})$.

F. Privacy-Preserving On-Device Webcam Proctoring

The proctoring module captures the candidate's camera stream via `getUserMedia` and runs a quantized MobileNet-SSD detector in-browser using TensorFlow.js over WebGL. No video leaves the client device. The detector monitors eight prohibited gadget classes:

$$G = \{\text{cell phone, tablet, laptop, tv, remote, keyboard, mouse, book}\} \quad (4)$$

To prevent false alarms from transient frame noise, IPMS applies a temporal persistence filter. Let $D_t(c)$ be the detection confidence for class c in G at frame t , sampled every Δt ms (default 1000 ms), with confidence threshold θ_{conf} and administrator allowlist A :

$$H_t(c) = \begin{cases} 1 & \text{if } D_t(c) \geq \theta_{\text{conf}} \wedge c \notin A \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

$$C_t(c) = \begin{cases} C_{t-1}(c) + 1 & \text{if } H_t(c) = 1 \\ 0 & \text{if } H_t(c) = 0 \end{cases} \quad (6)$$

An infraction is flagged only when consecutive positive hits satisfy $k_{\text{frames}} \in [1, 5]$:

$$\Psi_t(c) = \mathbb{I} \left(\text{Big}(C_t(c) \geq k_{\text{frames}}) \right) \quad (7)$$

G. Browser Confinement & Workspace Lockdown

During assessments, the HTML5 Fullscreen API is engaged. Capture-phase listeners intercept shortcut keys (Alt+Tab, Ctrl+C/V/X, F5, F12). Tab switches trigger `visibilitychange` events feeding a two-strike policy: a first violation issues an on-screen warning with a 5-second countdown; a second infraction enforces automatic submission.

H. ChatGPT-Style Conversational AI Assistant

The platform incorporates a unified conversational intelligence assistant featuring a frameless streaming markdown canvas, right-aligned user bubbles, text-to-speech audio synthesis, microphone speech-to-text dictation, deep reasoning reflection ("Think" mode), and collapsible session history. The assistant operates across three distinct security roles:

- **Admin Oracle:** Queries student statistics, drive progress, and audit logs.

- *Company Recruiter AI*: Summarizes candidate applicant performance and drive funnels.
- *Student Mentor*: Conducts mock interviews and provides technical roadmap guidance.

I. State Hydration & Formative Feedback

Test progress is continuously mirrored in `localStorage` scoped by candidate and test ID, enabling recovery in case of accidental page reload. Upon submission, a serverless function evaluates incorrect questions and delivers diagnostic remedial feedback referencing study topics.

VI. EMPIRICAL EVALUATION & EXPERIMENTAL RESULTS

To rigorously assess the system's operational viability, we conducted extensive experimental evaluations covering proctoring accuracy, document verification precision, latency throughput, test suite coverage, and user experience.

A. Experimental Setup

Experiments were conducted across three hardware tiers: (Tier 1) Apple MacBook Air M2 (8-core GPU), (Tier 2) Intel Core i5-1135G7 (Intel Iris Xe), and (Tier 3) Entry-level Celeron N4020 (Intel UHD 600). WebCam video was sampled at 1280×720 resolution under diverse illumination conditions.

B. Proctoring Pipeline Performance

We evaluated 500 controlled test sessions featuring deliberate cheating infractions (introducing smartphones, notebooks, second screens) and ordinary non-cheating gestures (drinking water, adjusting glasses, chin-resting). Table III presents the detection metrics across confidence thresholds θ_{conf} and persistence limits k_{frames} .

TABLE III
ON-DEVICE PROCTORING ACCURACY & CLIENT LATENCY

θ_{conf}	k_{frames}	True Pos. (TPR %)	False Pos. (FPR %)	Mean Latency	GPU Mem
0.50	1	99.4%	14.6%	28.4 ms	142 MB
0.60	2	98.6%	4.2%	29.1 ms	144 MB
0.65	3	97.8%	1.2%	31.2 ms	146 MB
0.75	3	94.1%	0.8%	31.8 ms	146 MB
0.80	4	88.5%	0.2%	32.6 ms	148 MB

The configuration $\theta_{\text{conf}} = 0.65$ with $k_{\text{frames}} = 3$ achieves an optimal balance, maintaining a 97.8% detection sensitivity while suppressing false alarms to 1.2%.

C. Multimodal Document Verification Precision

The document verification pipeline was evaluated using 120 institutional marks card samples across four formats: clean digital PDFs ($n=35$), high-resolution smartphone scans

($n=35$), perspective-skewed low-light captures ($n=30$), and digitally altered specimens ($n=20$). Results are tabulated in Table IV.

TABLE IV
MULTIMODAL MARKS CARD EXTRACTION & TAMPER SCREENING

Document Class	USN / Name Match	SGPA Accuracy	Tamper Detection	Mean Latency
Clean Digital PDF	100.0%	100.0%	N/A	1.12 s
Mobile Scan (High Res)	98.6%	97.8%	N/A	1.48 s
Perspective Skewed / Low Light	94.3%	93.1%	N/A	1.82 s
Digitally Altered Specimen	90.0%	92.5%	95.0% (19/20)	1.64 s
Aggregate Performance	96.4%	96.1%	95.0%	1.46 s

D. System Latency and Throughput Benchmarks

We benchmarked key operational workflows against conventional server-side batch operations across a cohort of 500 candidate profiles and 25 active company drives. Table V demonstrates the latency advantages achieved by client-evaluated reactive caching.

TABLE V
LATENCY COMPARISON: CONVENTIONAL BATCH VS. IPMS REAL-TIME

Operational Workflow	Conventional Batch	IPMS Client/Edge	Speedup Factor
Eligibility Match (500 candidates)	14,280 ms	4.2 ms	3,400×
Recruiter JD AI Extraction	Manual (15-30 min)	2,410 ms	>400×
AI Test Question Synthesis (10 Qs)	Manual (45 min)	3,850 ms	>700×
Drive Event Notification Relay	18,500 ms (Email batch)	85 ms (WebSocket)	217×
Exam State Crash Recovery	Unrecoverable / Zero	18 ms (Hydrated)	Instant

E. Automated Test Suite and Code Reliability

The IPMS codebase includes an automated test harness executed through Vitest and React Testing Library. The suite contains 57 tests structured across six domains, achieving 100% pass rate. Table VI details the testing coverage breakdown.

TABLE VI
AUTOMATED VITEST TEST SUITE COVERAGE
BREAKDOWN

Functional Test Suite	Test Count	Target Verification Scope	Status
Anti-Cheat & Confinement	9	Key interception, tab switch count, auto-submission	9 / 9 Passed
Eligibility Permutations	5	CGPA cutoffs, branch arrays, backlogs, skills	5 / 5 Passed
Scoring & Pass/Fail Logic	12	Boundary percentage evaluation, tie-breaking	12 / 12 Passed
Round State Machine	8	Candidate progression, gating, lockout deadlines	8 / 8 Passed
Auth, Session & Protected Routes	10	Role-based navigation, 2FA prompt, idle timeout	10 / 10 Passed
Theme & Utility Functions	13	Dark/Light persistence, contrast math, sanitization	13 / 13 Passed
Total Suite Verification	57	Comprehensive full-stack verification	57 / 57 Passed

F. Usability Evaluation (SUS Benchmark)

A structured user study was conducted with 54 participants: 40 engineering students, 8 Training and Placement coordinators, and 6 visiting corporate recruiters. Participants executed standard end-to-end tasks and evaluated the platform using the standard 10-item System Usability Scale (SUS) [12]. Table VII outlines the role-wise usability metrics.

TABLE VII
SYSTEM USABILITY SCALE (SUS) BENCHMARK
EVALUATION

User Cohort	Sample Size	Mean SUS Score	Standard Dev.	Adjective Rating
Student Candidates	40	87.5 / 100	± 4.8	Excellent (Grade A)
T&P Coordinators / Faculty	8	84.0 / 100	± 3.6	Good (Grade A-)
Corporate HR Recruiters	6	88.2 / 100	± 3.1	Excellent (Grade A)
Combined Benchmark	54	86.4 / 100	± 4.2	Grade A ("Exceptional")

VII. DISCUSSION

The primary architectural trade-off in IPMS is the choice of client-side machine learning for invigilation rather than server-side video streaming. Processing WebGL frames locally eliminates the recurring bandwidth cost of cloud video streaming—a critical barrier for students in rural connectivity regions. Furthermore, keeping webcam video isolated within browser GPU memory eliminates the data-privacy liabilities associated with transmitting biometric video feeds to cloud infrastructure [6]. The trade-off is the constraint of client GPU compute budgets; as shown in Section VI-B, the MobileNet-SSD detector utilizes ~146 MB of GPU RAM and achieves a 31.2 ms inference latency on mid-range hardware, which fits comfortably within standard interactive browser budgets.

Secondly, client-side evaluation of the eligibility formula (1) avoids server-side database locking and batch-synchronization

lag. Because the query cache holds student credentials and company parameters locally, filtering 500 candidates across 25 drives completes in 4.2 ms, compared to 14.2 seconds for legacy SQL table scans.

VIII. LIMITATIONS AND FUTURE SCOPE

While the platform fulfills core placement requirements, several extensions remain for future iterations:

- *Sandboxed Code Execution Engine*: Programming assessments are currently evaluated via output matching; integrating a sandboxed WebAssembly execution runtime (e.g., Pyodide / Wasmer) would enable testing against hidden test cases.
- *Facial Landmark Alignment*: While object detection captures unauthorized hardware, integrating on-device MediaPipe facial mesh models would enable verification of candidate head pose and multi-face presence without video egress.
- *Enterprise HRMS Synchronization*: Developing bi-directional webhooks with institutional Enterprise Resource Planning (ERP) systems would automate final offer acceptance reconciliation.

IX. CONCLUSION

This paper presented the architectural design, formal modeling, and empirical verification of the Intelligent Placement Management System (IPMS Elite). By unifying reactive eligibility evaluation, multimodal marks card OCR verification, in-browser privacy-preserving webcam invigilation, and end-to-end recruiter onboarding into a single relational ecosystem, IPMS eliminates the administrative fragmentation of traditional campus placement operations. Experimental benchmarks demonstrate high proctoring sensitivity (97.8% TPR at 1.2% FPR), rapid client-side eligibility evaluation (<5 ms), robust test coverage (57/57 passed tests), and exceptional user satisfaction (SUS score 86.4/100). IPMS offers an auditable, extensible reference implementation for higher education institutions modernizing their recruitment infrastructure.

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